

Macari Magdy Fayez Abdelmalak

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PROFILE SUMMARY

MSc Physics student at the University of Bologna specializing in computational condensed-matter physics. Experienced in density functional theory, electronic-structure calculations, topological materials, surface chemistry, and ab-initio molecular dynamics using VASP, Quantum ESPRESSO, Wannier90, and WannierTools. Seeking a research internship or thesis project in computational materials physics and quantum materials.

EDUCATION

2025–Present	MSc Physics, Alma Mater Studiorum – University of Bologna	Expected: 2027
2025–Present	Professional Master’s in Computer Science (Quantum Computing), Alexandria University	
2020–2024	BSc Physics, University of Science and Technology at Zewail City	
	Graduation Project: <i>Quantum Teleportation of Multiple Degrees of Freedom of a Single Photon Using Hyperentanglement</i> [Link]	June 2024
	Partial Merit-Based Scholarship	

SELECTED COMPUTATIONAL MATERIALS PROJECTS

Department of Physics and Astronomy “Augusto Righi”, University of Bologna

Bi₂Se₃ Topological Insulator: DFT and Wannier Analysis (2025)

- Performed VASP calculations with and without spin–orbit coupling, identifying band inversion and a bulk gap of ≈ 0.29 eV.
- Used Wannier90 and WannierTools to obtain surface Dirac states and Z_2 indices (1;000).

Au(111) Surface Chemistry and Ab-Initio Molecular Dynamics (2025)

- Performed Quantum ESPRESSO calculations of bulk Au, Au(111) surface energy, and H₂O/CH₃OH adsorption.
- Simulated water and methanol confined between Au(111) slabs using ab-initio molecular dynamics at 7 and 14 GPa.

TECHNICAL SKILLS

Electronic Structure: VASP, Quantum ESPRESSO, Wannier90, WannierTools, DFT, ab-initio molecular dynamics.

Programming and HPC: Python, C/C++, Bash, Linux/Unix, SLURM, Git/GitHub, LaTeX.

Analysis and Visualization: XCrySDen, Origin, CERN ROOT, Mathematica.

Quantum Computing: Qiskit, PennyLane, TensorFlow, Keras, Classiq.

EXPERIENCE

Physics Teaching Assistant (Feb–May 2025)

University of Science and Technology at Zewail City (*UST-ZC*)

- Led problem-solving sessions in electromagnetism and supported thermodynamics, waves, and optics laboratories.

Physics Teaching Assistant — School of Engineering (Sep–Dec 2024)

Misr University for Science and Technology (*MUST*)

- Led online and in-person problem-solving sessions in classical mechanics and modern physics.

INTERNSHIPS AND SCHOOLS

ICTP–IBM School on Quantum Computing and Simulation ICTP, Trieste, Italy (20–30 April 2026)

- Completed school on quantum computing and simulation.

Womanium Global Quantum + AI Program (2024)

- Ten-week program in quantum computing, quantum hardware, algorithms, and quantum machine learning.

3rd Summer School and Internship Programme at CTP British University in Egypt (*BUE*) (2023)

- Advanced-physics lectures; projects on alternative Λ CDM cosmology [\[Link\]](#) and Standard Model gauge bosons [\[Link\]](#).

AWARDS

Third Place — Quantum-AI Hackathon, Nile University (2024)

- Developed a quantum machine-learning classification model using Qiskit Machine Learning.

ONLINE COURSES

Machine Learning Specialization, Coursera (2023) [\[Certificate\]](#)

QxQ: Qubit by Qubit — Introduction to Quantum Computing and Algorithms, IBM (2024)

QClass23/24: Fundamentals of Quantum Computers with Programming, QWorld (2024)

LANGUAGES

Arabic: Native | English: Professional working proficiency (IELTS 6.5)